

Permutation Patterns

William Turner

Department of Mathematics & Computer Science

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Outline

- 1 Combinatorics
- 2 Permutations
- 3 Permutation Patterns
- 4 An Application

Combinatorics is concerned with the study of...

- arrangements
- patterns
- designs
- assignments
- schedules
- connections
- configurations

Combinatorical Problems

- A computer scientist considers *patterns* of digits and switches to encode complicated statements.
- A shop supervisor prepares *assignments* of workers to tools or to work areas.
- An agronomist *assigns* test crops to particular fields.
- An electrical engineer considers alternative *configurations* for a circuit.
- An industrial engineer considers alternative production *schedules* and workplace *configurations* to maximize efficient production.
- A university scheduling officer *arranges* class meeting times and students' *schedules*.
- A chemist considers possible *connections* between various atoms and molecules and *arrangements* of atoms into molecules.
- A transportation officer *arranges* bus or plane *schedules*.
- A linguist considers *arrangements* of words in unknown alphabets.

Three Basic Problems

Existence: Is there at least one arrangement of a particular kind?

Counting: How many arrangements are there?

Optimization: Among all possible arrangements, what is the best one according to some criterion?

Basic Counting Rules

The Sum Rule

If one event can occur in n_1 ways and a second event in n_2 different ways, there are $n_1 + n_2$ ways in which either the first event or the second event can occur, but not both.

Example

I have five apples and four oranges. How many ways can I select one piece of fruit—either an apple or an orange?

$$5 + 4 = 9$$

Basic Counting Rules

The Product Rule

If one event can occur in n_1 ways and a second event in n_2 different ways, independent of the first, then there are $n_1 \cdot n_2$ ways in which both the first event and the second event can occur.

Example

I have five apples and four oranges. How many ways can I select one apple and one orange??

$$5 \cdot 4 = 20$$

- A *permutation* is an arrangement of the numbers in the set

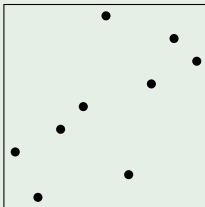
$$[n] = \{1, 2, 3, \dots, n\}$$

for some $n > 0$.

- The length of permutation σ is denoted $|\sigma|$.
- S_n is the set of all permutations of length n .
- Common to plot permutations graphically by points $(i, \sigma(i))$ where $\sigma = \sigma(1) \dots \sigma(n)$.

Example

Consider $\sigma = 314592687 \in S_9$.



Counting Permutations

How many permutations are there of length n ?

There are n ways to choose the first element, $n - 1$ ways to choose the second, ... , $n - i$ ways to choose the $i + 1$ -st element, ..., 1 way to choose the n th element:

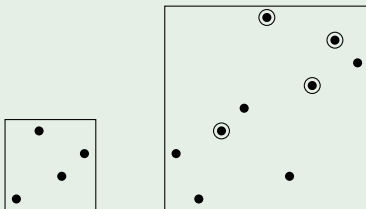
$$|S_n| = n(n - 1)(n - 2) \cdots (n - i) \cdots 1 = n!$$

Permutation Pattern

- Consider some subsequence (not necessarily contiguous) of σ .
- Replace i -smallest entry by i (Reduce their values)
- This forms a *pattern* π contained in σ ($\pi \leq \sigma$)

Example

$1423 \leq 314592687$ because 4968 (among others) is ordered the same way:

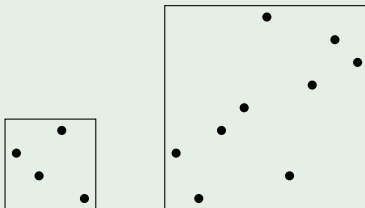


Pattern Avoidance

- σ avoids π if $\pi \not\leq \sigma$
- $\sigma \in S_n(\pi)$

Example

$314592687 \in S_9(3241)$ because $3241 \not\leq 314592687$:

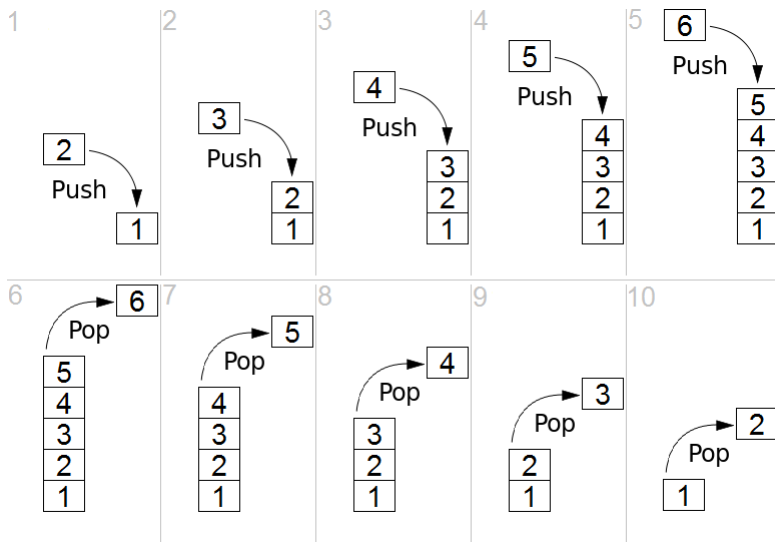


- Computer scientists want to know how to quickly sort a list (permutation).
- Knuth (1968): What permutations can be sorted by an algorithm with a single stack as storage?
- Linear time algorithm ($2n$ operations — fast!)

Stack

- Linear structure with insertion and deletion on same end
- “Push” and “pop”
- Last in, first out (LIFO)
- E.g., stack of plates in cafeteria

Stack Example



Sorting with a Stack

Knuth's algorithm

- Initialize an empty stack
- For each input value x :
 - While the stack is nonempty and x is larger than the top item on the stack, pop the stack to the output
 - Push x onto the stack
- While the stack is nonempty, pop it to the output

Example (4132)

- 1 Push 4 onto stack
- 2 Push 1 onto stack
- 3 Pop 1 off of stack
- 4 Push 3 onto stack

- 5 Push 2 onto stack
- 6 Pop 2 off of stack
- 7 Pop 3 off of stack
- 8 Pop 4 off of stack

Sorting with a Stack

Example (3142)

- | | |
|----------------------|----------------------|
| ① Push 3 onto stack | ⑤ Push 4 onto stack |
| ② Push 1 onto stack | ⑥ Push 2 onto stack |
| ③ Pop 1 off of stack | ⑦ Pop 2 off of stack |
| ④ Pop 3 off of stack | ⑧ Pop 4 off of stack |

- This produces 1324
- The problem is the 342, or the pattern 231



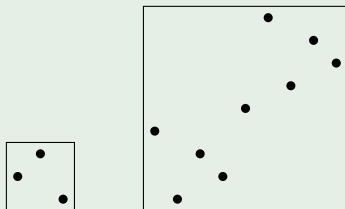
Stack Sortable

Theorem

A permutation σ is stack sortable if and only if $\sigma \in S_n(231)$.

Example

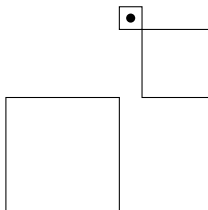
The permutation 413259687 is stack sortable.



Counting Stack Sortable Permutations

How big is $|S_n(231)|$?

$\sigma \in S_n(231)$ if and only if σ has recursive structure



If $|S_n(231)| = c_n$, then $c_n = \sum_{i=1}^n c_{i-1} c_{n-i}$.

Counting Stack Sortable Permutations

Using some standard methods, we find

$$c_n = \frac{1}{n+1} \binom{2n}{n} = \frac{(2n)!}{(n+1)!n!} = C_n,$$

the *Catalan numbers*, which count many combinatorial objects:

- [Wikipedia](#)
- [The Online Encyclopedia of Integer Sequences](#)